

PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Metro Ticket Generating System

Date	Team ID	Project Name	Maximum Marks
04 September 2026	_____	Metro Ticket Generating System	4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
1	Metro Ticket Booking	Access Book a Metro Ticket catalog item; select Starting From and Going To; select journey type; enter number of passengers.
2	Automatic Fare Calculation	Calculate and populate the applicable fare based on selected route, journey type and passenger count.
3	Payment Mode	Select a payment mode such as UPI, Card or Others; display the additional payment-mode field when Others is selected.
4	Form Validation & Data Integrity	Enforce mandatory/non-empty fields, validate source/destination and fare matching, and prevent incomplete submissions.
5	Ticket Request Creation	Submit the completed catalog form and create the corresponding ServiceNow request / requested item record.
6	QR Ticket Generation	Generate a QR-code URL from the created ticket/request identifier and present the QR ticket after submission.
7	Service Portal QR Display	Display the generated QR code through the Metro QR Widget in the Service Portal.
8	Access Control	Allow customers/users to use the booking experience while restricting direct backend database-table access according to configured ACLs.
9	Automation & Field Mapping	Use Catalog Client Scripts, UI Policies and process automation to populate fields, control visibility and map submitted values.
10	Testing & Verification	Support testing of form behaviour, fare calculation, QR generation, visibility and data-integrity conditions.

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

No.	Non-Functional Requirement	Description
1	Usability	The booking form should guide passengers through route, journey, passenger and payment selections with clear field behaviour and conditional visibility.
2	Security	Role-based ACLs should protect backend ticket data so intended users can book tickets without direct access to restricted database tables.
3	Reliability	Fare calculation, mandatory-field validation, field mapping and QR-generation logic should produce consistent results for valid booking inputs.
4	Performance	The booking form, client-side calculations and QR-generation interaction should respond without unnecessary manual processing or repeated data entry.
5	Availability	The ServiceNow catalog/portal booking experience should be available to authorized users whenever the ServiceNow instance is available.
6	Scalability	The design should allow additional metro stations, routes, fares, journey options and future integrations to be added without redesigning the entire booking workflow.